# Statistics in the turbulence talk figures

Formulas recovered from `codes/talk_figures.py`, `codes/paper_figures_li.py`, `turbulence/paper_figs_ensemble.py`, `codes/check_moments.py`.
Each formula is followed by its LaTeX code (copy from the grey box; math mode, no custom macros). Grey *Code:* lines are implementation details, not for the slides.

## Notation (common to all figures)

$N$ signals $x_b$, $b = 1, \ldots, N$, each of $T$ samples. Lags $\tau$ are integers (samples). Lagrangian figures show the lag as $\tau/\tau_\eta$ ($\tau_\eta$: Kolmogorov time in samples); Eulerian (jets) figures show $\tau$ in samples.

$$\delta_\tau x(t) = x(t+\tau) - x(t)$$

```
\delta_\tau x(t) = x(t+\tau) - x(t)
```

$\langle\cdot\rangle$: average over all signals and over all times $t$ at which the quantity exists. $\sigma$: standard deviation,

$$\sigma(y)^2 = \langle y^2 \rangle - \langle y \rangle^2$$

```
\sigma(y)^2 = \langle y^2 \rangle - \langle y \rangle^2
```

## Structure functions $S_p(\tau)$

Figures: `li_structure_functions` ($p = 2, 4, 6, 8$, Lagrangian); `fig6_*_1`, right panel ($p = 4, 6, 8$, both). Curves shifted down by $10^{-1}$ per order.

$$S_p(\tau) = \left\langle (\delta_\tau x)^p \right\rangle$$

```
S_p(\tau) = \big\langle (\delta_\tau x)^p \big\rangle
```

*Code:* Fig. 6 takes the absolute value of the mean, $|\langle(\delta_\tau x)^p\rangle|$: identical for even $p$. Lags: $\tau = 1, \ldots, T/2 - 1$ (Fig. 6), $\tau = 1, \ldots, \min(\texttt{max_lag}, T - 2)$ (Li).

## Flatness $F_p(\tau)$

Figure: `li_flatness` ($p = 4, 6, 8$, Lagrangian).

$$F_p(\tau) = \frac{S_p(\tau)}{S_2(\tau)^{p/2}}$$

```
F_p(\tau) = \frac{S_p(\tau)}{S_2(\tau)^{p/2}}
```

Gaussian values, for reference (not drawn in the figure):

$$F_4 = 3, \qquad F_6 = 15, \qquad F_8 = 105$$

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### Flatness slope

Figure: `fig6_*_2`, left panel (both). Dotted line at 0.

$$\frac{d \log F_4}{d \log \tau}$$

```
\frac{d \log F_4}{d \log \tau}
```

*Code:* Evaluated on 30 log-spaced lags in $[1, T/2)$, centred second-order finite differences in $\log \tau$ (`np.gradient`).

### Local scaling exponent $\zeta_p(\tau)$

Figure: `li_local_exponents` ($p = 4, 6, 8$, Lagrangian). Dotted line at $p/2$ (no intermittency: $S_p \propto S_2^{p/2}$).

$$\zeta_p(\tau) = \frac{d \log S_p}{d \log S_2}$$

```
\zeta_p(\tau) = \frac{d \log S_p}{d \log S_2}
```

*Code:* Computed as the ratio $\frac{d \log S_p / d \log \tau}{d \log S_2 / d \log \tau}$ (same thing, chain rule), finite differences on the unit-spaced lag grid $\tau = 1, 2, \ldots$, as Li et al. 2024.

### Increment PDFs

$$\mathrm{PDF}\left( \frac{\delta_\tau x}{\sigma(\delta_\tau x)} \right)$$

```
\mathrm{PDF}\!\left( \frac{\delta_\tau x}{\sigma(\delta_\tau x)} \right)
```

`li_increment_pdfs` (Lagrangian): normalised density, $\tau/\tau_\eta = 1, 2, 5$, shifted down by $10^{-3}$ per lag. `fig6_*_2`, middle panel (both): PDF divided by its maximum (peak = 1), $\tau = 1, 4, 16, 64$ samples, shifted down by $10^{-1}$ per lag.

*Code:* Fig. 6: histogram on $[-6, 6]$, 201 bin edges, Gaussian smoothing of 1.5 bins. Li: no smoothing, bins over $\pm \max |\delta_\tau x|/\sigma$.

### Marginal PDF

Figure: `fig6_*_1`, left panel (both).

$$\mathrm{PDF}\left( \frac{x}{\sigma(x)} \right)$$

```
\mathrm{PDF}\!\left( \frac{x}{\sigma(x)} \right)
```

*Code:* The data are standardised ($\langle x \rangle = 0$, $\sigma(x) = 1$), so the code histograms $x$ directly; the model samples are put on the same bins without re-standardising. Gaussian smoothing of 1.5 bins.

### Power spectrum $P(\omega)$

Figure: `fig6_*_1`, middle panel (both); $\omega = k/T$ ($k = 1, \ldots, T/2$) in cycles per sample, shown for $0 < \omega \leq 0.4$.

$$P(\omega) = \frac{1}{N} \sum_{b=1}^{N} \Big| \sum_{t=0}^{T-1} x_b(t)\, e^{-2\pi i \omega t} \Big|^2$$

```
P(\omega) = \frac{1}{N} \sum_{b=1}^{N} \Big| \sum_{t=0}^{T-1} x_b(t)\, e^{-2\pi i \omega
    t} \Big|^2
```

*Code:* No $1/T$ normalisation (`|rfft|`$^2$ averaged over signals).

### Skewness

Figure: `fig6_*_2`, right panel (both). Dotted line at 0.

$$\text{skewness}(y) = \frac{\big\langle \big(y - \langle y\rangle\big)^3 \big\rangle}{\sigma(y)^3}, \qquad y = x(t)\,\delta_\tau x(t)$$

```
\mathrm{skewness}(y) = \frac{\big\langle \big(y - \langle y \rangle\big)^3
    \big\rangle}{\sigma(y)^3}, \qquad y = x(t)\,\delta_\tau x(t)
```

*Code:* `skew_mode='power'` (the talk setting). 45 log-spaced lags in $[1, T/2)$.

### Cross-scale correlations $C_{p,q}(\tau, \tau^\star)$

Figure: `fig6_*_3` (both), $\tau^\star = 20$ samples (dotted vertical line; shown as $\tau^\star/\tau_\eta$ in the Lagrangian figure). Horizontal line at 1: value when the increment amplitudes at the two scales are uncorrelated.

$$C_{p,q}(\tau, \tau^\star) = \frac{\big\langle |\delta_{\tau^\star} x|^p \, |\delta_\tau x|^q \big\rangle}{\big\langle |\delta_{\tau^\star} x|^p \big\rangle \, \big\langle |\delta_\tau x|^q \big\rangle}$$

```
C_{p,q}(\tau, \tau^\star) = \frac{\big\langle |\delta_{\tau^\star} x|^p \, |\delta_\tau
    x|^q \big\rangle}{\big\langle |\delta_{\tau^\star} x|^p \big\rangle \, \big\langle
    |\delta_\tau x|^q \big\rangle}
```

The three panels are $C_4 \equiv C_{2,2}$, $C_6 \equiv C_{3,3}$ and $C_{4,2}$. For even $p, q$ the absolute values are not needed, e.g.

$$C_4(\tau, \tau^\star) = \frac{\big\langle (\delta_{\tau^\star} x)^2 \, (\delta_\tau x)^2 \big\rangle}{S_2(\tau^\star)\, S_2(\tau)}$$

```
C_4(\tau, \tau^\star) = \frac{\big\langle (\delta_{\tau^\star} x)^2 \, (\delta_\tau x)^2
    \big\rangle}{S_2(\tau^\star)\, S_2(\tau)}
```

*Code:* Both increments start at the same time $t$; the numerator averages over $t \le T - 1 - \max(\tau, \tau^\star)$, each denominator factor over its own full range. $10^{-8}$ added to the denominator. Lags $\tau = 1, \ldots, T/2 - 1$.

### Rare signals: acceleration criterion

Acceleration = one-step increment (as Li et al.):

$$a(t) = \delta_1 x(t) = x(t+1) - x(t)$$

```
a(t) = \delta_1 x(t) = x(t+1) - x(t)
```

Each signal is summarised by its peak acceleration $\max_t |a(t)|$. **Criterion of the talk**: a signal is rare if

$$\max_t |a(t)| \;>\; \text{percentile}_{95}\big(\max_t |a|\big)$$

```
\max_t |a(t)| \;>\; \mathrm{percentile}_{95}\big( \max_t |a| \big)
```

where the percentile is taken over the data signals: rare = the 5 % of the data signals with the largest peak acceleration.

**Alternative** (`turb_rare_typical_sigma.ipynb`, threshold in units of the standard deviation of $a$): rare if

$$\max_t |a(t)| \;\geq\; 20\,\sigma(a)$$

```
\max_t |a(t)| \;\ge\; 20\,\sigma(a)
```

where $\sigma(a)$ is computed over every time of every data signal. One single number, so the threshold is the same level for the data and for the generated signals.

*Code:* `rare_events_data_vs_generated` applies the same rule at the lags $\tau/\tau_\eta = 20, 5, 1$, with $a$ replaced by $\delta_\tau x$ and the percentile taken at that lag on the data; the generated signals use the same absolute threshold. `rare_events_by_logp` instead picks signals among the 1 % with the highest $-\log p$.

### $-\log p$: what the mean and the std are taken over

The model is $p(x) \propto \exp(+\Theta \cdot \phi(x))$ ($\Theta = \Theta_{\text{reg}}$ at the last time step), so

$$-\log p(x) = -\,\Theta \cdot \phi(x) + \text{const}$$

```
-\log p(x) = -\,\Theta \cdot \phi(x) + \mathrm{const}
```

The constant is unknown, so the level of $-\log p$ means nothing: only differences between signals do. Large $-\log p$ = unlikely signal (rare signals sit on the right of the histograms).

**Mean and std are over signals, for one fixed $\Theta$**: $x_b$, $b = 1, \ldots, N$, are the $N = 10000$ data signals (the training signals):

$$\text{mean} = \frac{1}{N}\sum_{b=1}^{N}\big[-\log p(x_b)\big], \qquad \text{std} = \left(\frac{1}{N-1}\sum_{b=1}^{N}\Big[-\log p(x_b) - \text{mean}\Big]^2\right)^{1/2}$$

```
\mathrm{mean} = \frac{1}{N} \sum_{b=1}^{N} \big[-\log p(x_b)\big], \qquad \mathrm{std} =
    \Bigg( \frac{1}{N-1} \sum_{b=1}^{N} \Big[-\log p(x_b) - \mathrm{mean}\Big]^2
    \Bigg)^{1/2}
```

- **Table** (per category $c$ of potentials): the same mean and std, applied to the contribution of the category $-\sum_{F\in c}\Theta_F \cdot \phi_F(x_b)$ instead of $-\log p(x_b)$. Computed for each seed ($\Theta$ differs from seed to seed, the data signals are the same), then *averaged over the seeds*.
- **Bar plots**: same quantities; error bars = std over seeds. Third panel: std of the category divided by the std of $-\log p$.
- **Histograms**: $-\log p(x_b) -$ median, the median taken over the signals shown (all data signals, or all generated signals, each with its own median).
- **Gaussian test**: $-\log p(x_b) -$ mean, the mean taken over the samples of its own law (training signals for the exact $-\log p$, generated signals for the MGD one). For $M$ dimensions the exact $-\log p = \chi^2_M/2 + \text{const}$, hence std $= \sqrt{M/2}$.

*Code:* The sign flip leaves every std unchanged and changes the sign of every mean. Category contributions add up to $-\log p + \text{const}$, but their stds do not add up (covariances between categories). The mean of a category contribution is a level only: a constant added to $\phi_F$ shifts it without changing $p$. Code: `ddof=1` for every std; `energy_table`, `plot_energy_bars`, `plot_logp_histogram`, `plot_logp_gaussian` in `codes/talk_figures.py`.

### Error bands (all figures)

Line: the statistic on all signals. Band, `band='batch'` (default): min/max of the same statistic over 10 disjoint batches of signals (data and model, as Li et al.). `band='seeds'`: model line = mean over independent runs, band = $\pm 1$ std; the data band stays min/max over batches.